Assignment 1

Question 1. The Mathematics of Qubits

Consider the states

$$|+\rangle = \frac{|0\rangle + |1\rangle}{\sqrt{2}}, \qquad |-\rangle = \frac{|0\rangle - |1\rangle}{\sqrt{2}}$$

Recall that Bloch sphere representation for qubits:

$$|\psi\rangle = \cos\frac{\theta}{2}|0\rangle + e^{i\phi}\sin\frac{\theta}{2}|1\rangle$$

where $\theta \in [0, \pi]$ is polar angle and $\phi \in [0,2\pi)$ is azimuthal angle.

a. **(4 points)** Write $|+\rangle$ and $|-\rangle$ in matrix (column vector) form. Write down their corresponding bra vectors $\langle +|$ and $\langle -|$.
b. **(4 points)** Find Bloch sphere angles (θ, ϕ) for each of $|+\rangle$ and $|-\rangle$.
c. **(2 points)** Plot $|+\rangle$ and $|-\rangle$ on the Bloch sphere (you may use an online Bloch sphere visualizer).
d. **(4 points)** Compute probability of measuring $|0\rangle$ and $|1\rangle$ when in each of the states $|+\rangle$ and $|-\rangle$.
e. **(12 points)** Apply each of the Pauli matrices $\hat{X} = \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}, \hat{Y} = \begin{bmatrix} 0 & -i \\ i & 0 \end{bmatrix}, \hat{Z} = \begin{bmatrix} 1 & 0 \\ 0 & -1 \end{bmatrix}$ to each of $|+\rangle$ and $|-\rangle$ and express results in $|+\rangle, |-\rangle$ basis.
f. **(4 points)** Show that $|+\rangle$ and $|-\rangle$ are orthogonal and normalized. Note: This mean that they form an orthonormal basis set.

Question 2. The Ramsey Sequence

Consider a two level system initialized at $|0\rangle$ with transition frequency ω_0. The detuning is $\delta\omega = \omega - \omega_0$ and the accumulated phase in time T is $\phi = \delta\omega \cdot T$.

a. **(4 points)** Apply a $\frac{\pi}{2}$ pulse to $|0\rangle$ and write down resulting state $|\psi_1\rangle$ in $|0\rangle, |1\rangle$ basis.
b. **(2 points)** Apply free precession operator for time T to $|\psi_1\rangle$ to obtain $|\psi_2\rangle$.
c. **(2 points)** Apply second $\frac{\pi}{2}$ pulse to $|\psi_2\rangle$ to obtain final state $|\psi_3\rangle$.
d. **(6 points)** Show that the probability of measuring $|1\rangle$ is

$$P_{|1\rangle} = \cos^2\frac{\phi}{2} = \cos^2\left(\frac{\delta\omega \cdot T}{2}\right)$$

e. **(2 points)** Plot this probability as a function of $\delta\omega$ for fixed T (you may use a plotting software).
    i. **(2 points)** How does $P_{|1\rangle}$ change when T is doubled?
    ii. **(2 points)** At what value of $\delta\omega$ is $P_{|1\rangle}$ maximized?

You may use the following:

- $\frac{\pi}{2}$ pulse about x: $R_x\left(\frac{\pi}{2}\right) = \frac{1}{\sqrt{2}}\begin{bmatrix}1 & i\\ i & 1\end{bmatrix}$
- Free precession about z: $U_{\text{free}}(T) = \begin{bmatrix}1 & 0\\ 0 & e^{i\phi}\end{bmatrix}$
- Trigonometry identities:

$$e^{i\phi} = \cos\phi + i\sin\phi$$
$$\cos^2\phi = \frac{1+\cos 2\phi}{2}$$
$$\sin^2\phi = \frac{1-\cos 2\phi}{2}$$